



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

PawPal: An Android-Based Smart Pet Feeder for Pet Owners in Saint Joseph Village 10

A Capstone Project Presented to the Faculty of the
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Pamantasan ng Lungsod ng Muntinlupa

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PAMANTASAN NG LUNGSOD NG MUNTINLUPA

TABLE OF CONTENTS

CHAPTER 1

INTRODUCTION

- a. Project Context
- b. Purpose and Description
- c. Objective of the Study
- d. Scope and Limitations

CHAPTER 2

REVIEW OF THE RELATED LITERATURE & STUDIES

- e. Technical Background
- f. Related Literature
- g. Related Studies
- h. Definition of Terms

CHAPTER 3

METHODOLOGY

- i. Requirement Analysis
- j. Requirement Documentation
- k. System Architecture
- l. Design of Software, Systems, Product and/or Processes
- m. Development and Testing
- n. Implementation Plan

CHAPTER 4

RESULTS & DISCUSSION

- a. Project Description
- b. Project Structures
- c. Project Capabilities and Limitation
- d. Project Evaluation

CHAPTER 5

SUMMARY OF FINDING, CONCLUSION & RECOMMENDATION

- a. Summary of Findings
- b. Conclusion
- c. Recommendation



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

Chapter 1 INTRODUCTION

PROJECT CONTEXT

The primary beneficiaries of PawPal: An Android-Based Smart Pet Feeder are pet owners in Saint Joseph Village 10 - Phase 1, a community with a high number of busy pet owners. These individuals often face challenges in maintaining consistent feeding schedules for their pets due to demanding work schedules, academic responsibilities, or other commitments. This inconsistency can negatively affect the health and well-being of pets, leading to underfeeding, overfeeding, or irregular dietary routines.

By addressing the needs of these pet owners and their pets, the project aims to enhance the quality of life for both. Pets will benefit from improved feeding habits and health, while owners gain peace of mind knowing their pets' nutritional needs are met efficiently, even during their absence. The PawPal system offers a technologically advanced and convenient solution, ensuring pets are properly nourished through automated and remotely monitored feeding schedules.

Many pet owners often encounter major problems when it comes to the daily eating patterns of their pets, particularly in ensuring that their pets eat on a



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

regular schedule with the proper portion of food in time. As the pet owners tend to engage in demanding work schedules, academic engagements or basic daily plans, irregular meal routines or in most cases, underfeeding or overfeeding occurs. Pet owners have the tendency of leaving a certain amount of food which can last for days or depending on how long they will be gone, it can be adequate or it may not be enough which can result in their pets either eating too much or not eating enough per day. Same scenarios occur if the owner happens to be out of town or in a distance far enough to keep them from monitoring their pets' dietary plans.

As a researcher and member of the community, the proponent recognizes the significance of addressing this problem. Pets are valued members of families, and their well-being directly impacts the quality of life and their availability on a daily set up. By developing an Android-based smart pet feeder tailored to the needs of pet owners Saint Joseph Village 10-Phase 1, the pet feeder aims to provide a solution to aid the owners from worrying and in ensuring their pets eating schedule were met.

The proposed solution, PawPal offers a technologically advanced approach to pet care. This Android-based smart pet feeder allows pet owners to schedule and monitor feeding times remotely through a mobile application. With customizable features like a notification, food portion, and an automatic feeding mechanics, PawPal ensures that pets receive proper nourishment at regular



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

schedules by dispensing a certain amount of dry food to their pets on a specified time and day. This will help owners multitask, especially if their commitments demand most of their time and may not be able to be attended to by their owners physically to make sure that they receive the nourishment they need.

PURPOSE AND DESCRIPTION

The purpose of this research is to develop an Android-based smart pet feeder that addresses the challenges of maintaining consistent feeding schedules for pets. This project aims to provide a practical and efficient solution that ensures proper nutrition and reduces the stress associated with pet care for owners with demanding lifestyles.

The proposed system, PawPal, integrates Internet of Things (IoT) technology with Arduino-based hardware to automate feeding routines. The NodeMCU 8266 will serve as the core component, managing food dispensing, scheduling, and monitoring functionalities. Pet owners can use the accompanying mobile application to remotely set feeding schedules and receive notifications about their pet's meals and food container levels.

By leveraging this innovative approach, the project seeks to improve the well-being of pets and provide convenience for their owners, ensuring reliable, automated feeding routines without requiring constant manual intervention.



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

OBJECTIVE OF THE STUDY

General Objective:

The main objective of this study is "Developing an Android-based smart pet feeder". It allows pet owners to schedule even when they are away from their home. Having a joyful and healthy pet at home relieves our stress. Creating a system that ensures your pets grow healthy while requiring little effort on your part is also beneficial.

Specific Objectives

1. To design an Android-based automatic pet feeder that uses hardware components for the following features:
 - a. Dispensing dry pet food on specific measurement.
 - b. Scheduling a meal three times a day (e.g breakfast, lunch, dinner).
 - c. Sending notifications to the owners once the food container is low.
 - d. Scheduling food plans using a mobile application that can be programmed for days, weeks and months.
 - e. Status notification of the food container.
2. Develop an Android-based automatic pet feeder using Arduino.



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

3. Provide automated reports everytime the pets have been fed including the feeding history.

SCOPE AND LIMITATION

This project focuses on creating a smart pet feeder for pet owners living in Saint Joseph Village 10 Phase 1, Langgam San Pedro Laguna. The feeder is designed to make pet care easier by automatically dispensing food, and letting owners set feeding schedules through an Android mobile app. The goal is to help busy pet owners take better care of their pets, especially when they are not home. By focusing on the specific needs of pet owners in this community, the project aims to provide a practical and useful solution.

However, there are some limitations to this project. The feeder will only work with common household pets like cats and dogs and is designed for dry food only. It will not support wet food or animals with special diets, such as exotic pets. Also, the mobile app can only be used on certain Android devices, so it may not be compatible with all phones. The feeder also needs a steady power supply to work properly. If there is a power outage, the feeding schedule may be disrupted. While the feeder can still be operated manually during these times, some features like remote control and notifications will not be available without internet access. Despite these limitations, the project is designed to



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

address the most common challenges faced by pet owners and provide a reliable and convenient solution for feeding their pets.

Chapter 2 REVIEW OF THE RELATED LITERATURE AND STUDIES

TECHNICAL BACKGROUND

Pet care technology has improved significantly with the development of an automatic smart pet feeder, combining numerous hardware components and software systems to create a convenient and efficient feeding solution for pets and their owners. This system uses NodeMCU , sensors, wireless communication modules, and real-time and date features to ensure perfect feeding schedules, manual accessibility, and remote accessibility. NodeMCU microcontroller functions as the smart pet feeder's central processing unit, coordinating the interactions between various parts. The NodeMCU , equipped with the ESP8266 chipset, provides an adaptable platform that may be used to read inputs, control outputs, and carry out programmed logic to efficiently automate feeding processes. By showing the current time and date as a visual feedback, improving user experience and ensuring user-friendly functionality.

A servo motor powers the feeding mechanism, rotating the food container and dispensing food into the pet's bowl. The servo motor provides automated feeding and dependable food delivery by transforming electrical signals into precise mechanical action, meeting the nutritional requirements and feeding



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

preferences of the pet. The real time and date interface gives precise time and date information, the feeder may follow feeding schedules that are preprogrammed and that dynamically adjust feeding times. Using the real-time clock interface, pet owners can adjust feeding schedules and intervals according to their pet's needs. To enable wireless communication and remote access, an ESP8266 NodeMCU module is integrated. The smart pet feeder can connect to local Wi-Fi networks and communicate with other devices, such laptops or smartphones, using the open-source development kit and firmware. Using a companion mobile application or web interface, pet owners may remotely monitor feeding activities, receive notifications, and modify settings, adding convenience and flexibility.

The level of food in the pet's bowl is also determined by the Ultrasonic Sensor. The Ultrasonic Sensor determines the distance between the food surface and the sensor in order to provide real-time feedback on food consumption, guaranteeing that the pet always has access to a sufficient amount of food. In order to improve the system's alerting capabilities, the Serial Monitor is instrumental. One such part is the buzzer, which is essential for alerting the pet owner to system alarms or providing a warning when the food container is empty. By having this option, the pet owner may be sure they are informed and able to react quickly to their requirements. The RFID reader is another crucial component of the system that offers manual feeding choices.



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

Through the use of RFID tags or cards, pet owners can manually start feeding when needed, providing flexibility in feeding schedules and ensuring that the pet eats on time even in unexpected situations.

In conclusion, the automated smart pet feeder—which is driven by a NodeMCU and a number of additional parts—represents an advanced approach to managing pet feeding. This system improves convenience and pet care by combining innovative sensors, wireless connectivity, and real-time scheduling and notification features. It also gives pet owners peace of mind and maintains the health and comfort of their cherished animals.

RELATED LITERATURE

Feeding is the most occurring problem pet owners face on a daily basis. Due to their daily engagements and busy schedules, their pets are often left alone for a day or two because of the activities the owners need to attend to. Some require them to stay out for long periods of time resulting in their pets being alone for the time being. [1]

Pets are considered no less than a family. They became a figure of being our emotional support and happy pill. Pets are like humans too, they need to be well taken care of and be fed regularly which is often the struggles of the owners. It is not easy to maintain a pet with today's busy lifestyle especially on their dietary aspects if the owners are always on the go and often out of the house to meet their daily busy schedules. Not all pet owners can feed their pets



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

according to a strict diet schedule. Many who work or study late into the day often miss feeding times. This problem is often underestimated, as some pet owners try to solve it by overfilling the food dish, which is not a safe solution. This approach is taken not only by busy pet owners but also by those who find feeding their pets three times a day to be a burdensome task. It's clear that pets need special care and attention, but a busy lifestyle makes it challenging to provide this care.[2]

Owning a pet is a widespread practice globally, and ensuring their well-being is ingrained in our human culture. While having a pet can offer emotional support, it also presents challenges. Numerous pet owners encounter significant issues when adopting and caring for their pets. Addressing such technological challenges requires embracing new innovative technologies, with IoT-based solutions emerging as a prominent option in this regard. [3]

Food is important for pets, like cats, just as it is for us. Usually, pet owners feed their pets by being at home and giving them food regularly. But sometimes, pet owners have to be away from home for a long time. In such cases, it's helpful to have a device that can automatically give food to pets, especially dry food. This way, pets can still be fed even when their owners are not at home and don't have extra time. [4]

In contemporary society, it is commonplace for households to include pets, necessitating attentive care. Ensuring timely meal provision for these animals is crucial, given their integral role within the family unit. Nevertheless, the demands



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

of modern life often lead some individuals to neglect their pets' feeding schedules due to time constraints. [5]

The pet care sector is growing quickly and accounting for a considerable amount of retail activity in the United States. Americans have significantly increased their pet spending over the last ten years, and this trend is predicted to continue in the years to come. The widespread problem of one pet eating another's food is still unresolved, even with the abundance of pet care items on the market. The advent of the Internet of Things (IoT) Smart Pet Feeder is a viable solution to this problem, benefiting both owners and their pets. Modern society is characterised by a general desire to own pets, which is explained by the belief that these animals can reduce stress, boredom, and loneliness. [6]

Particularly among older folks, people's perceptions of pets as significant components of their lives and activities are growing. Dogs and cats are the most popular pets, with many treating them like members of the family. The animal care sector is expanding along with the number of pets owned, which is driving the creation of new goods and innovations targeted at senior pet owners. [7]

Having pets can make life better, but it also means having duties that might stress people out. Pet owners understand that feeding their pets makes them happy and healthy, but sometimes they're too busy to do it, or they're away from home and can't check if their pets are eating. That's where a pet feeder can help. [8]



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

Feeding is the most common issue that pet owners deal with. Some pet owners leave their animals alone for a few days while they attend to other activities, training, or business matters outside the home, where the animals are not permitted to enter together due to their hectic daily schedules at work. Consequently, the automatic cat feeder was developed to enable pets to be fed continuously even when their owners are not present. [9]

Most of us own pets at home; some of us are patient enough to feed them on a daily basis, while others are not. Therefore, one of the biggest issues with pet ownership is finding time to care for them amid our hectic schedules. Even when we are gone from home for a few days, we still have to care for them. [10]

Some people search for the newest developments to make taking care of their pets at home easier because they have busy lives. This challenge shows us how important it is to create a remote-control feeding gadget so that pet owners may feed their animals more easily and maintain the natural growth of pets. [11]

Pet owners can choose to personally feed their animals or have pet feeders do it for them automatically. Solving the problems of pet owners' absence and feeding hygiene are the primary considerations here. Pet owners may feed their animals while they are gone thanks to the created automated method. [12]



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

RELATED STUDIES

According to the study of B. Rohitha, V. Divya Sree, B. Nikitha, and K. Pavan Kumar (2024), the pet feeder is a very good solution for pet owners who are not able to monitor their pets' feeding time. In their study, they proposed a pet feeder that doesn't only dispense food but also calls the pet during meal time. A push button is also included to control the amount of food to be dispensed onto the feeding bowl with the assistance of the Blynk mobile application regardless of the location of the owner. [13]

In 2023, J. Kulaikar, D. Kurade, A. Sawant, P. Sthawarmath, and A. Chaurasia published a study which is an IoT Based Automatic Pet Feeder and Monitoring System. The project includes Raspberry Pi as their module and having a monitoring system using a camera. They also integrated an automated feeding including water dispensing using a pump, it provided pets with a complete set of meals with the combination of water and dry pet food. [14]

In 2022, S. N. Mohamad, N. H. Mat Tahir, A. H. Marzuki, F. H. Azimi, S. R. Aw, W. F. W. Tarmizi, and M. L. Muhd Zain finished a study of an IoT based Real time Cat feeder in 2022. They used Arduino Uno as the microcontroller with an integration of RCT module using a Raspberry Pi model, Ultrasonic sensor responsible for detecting the distance of the pet from the feeder before the feeding actions execute and the servo motor for the output process making the open and close of the lid to finish the feeding process. [16]



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

Based on the study conducted by Safuan Naim Mohamad, Nurul Huda Mat Tahir, Ahmad Hakimi Marzuki, Farah Hanan Azimi, Suzanna Ridzwan Aw, and Wan Faizura Wan Tarmizi in (2022), they proposed a project that combines an Arduino Uno microcontroller, a servo motor for output, and two main components: an ultrasonic sensor and a real-time clock (RTC). Like radar and sonar systems, ultrasonic sensors evaluate reflected signals to analyse targets. This allows the object's distance to be measured by measuring the time interval between the signal's transmission and the reception of its echo. The RTC precisely times the feeding alerts to the Arduino, and the ultrasonic sensor measures the amount of food in the bowl. [17]

In 2021, Ayush Kumar Jaiswal(2021), proposed a system that uses the Internet of Things (IoT) pet feeder system using a dynamic control strategy. In accordance with the suggested strategy, users first determine the feeding time by considering their pet's needs. Through a user interface platform, users receive a smartphone alert at the appointed feeding time asking them to approve pet feeding. Additionally, users can choose to delay or refuse feeding during that period.

Users designate how many food doses to deliver once they have been given authorization. The Wi-Fi Module receives this data after that. Then, the Arduino board receives a signal from the Passive Infrared (PIR) sensor indicating that the pet is there. The motor is triggered by this signal to dispense the designated amount of food. [18]



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

According to the study of Muhammad Ayat Hidayat¹, Suriandi Jayakrista [2023], the Arduino microcontroller-based smart pet feeder system operates on a 9V power source, which initiates the movement of the servo to lower the food gate and activate essential components like load cells, ultrasonic sensors, buzzers, and LCDs. Directly connected to the microcontroller, the RTC (Real-Time Clock) serves as a scheduler for cat feeding times. Ultrasonic sensors gauge the remaining volume of cat feed, while the buzzer alerts when the feed is depleted, and the LCD displays feeding schedule information and remaining feed in the FoodyPet pet feeder. When the cat steps on the load cell, signaling the microcontroller, the servo motor is prompted to open the feed gate, allowing the cat immediate access to food without the need for conventional feeding by the owner. [19]

In 2022, Pushpa Birha, Ruchika Ingle, Samruddhi Tajne, Payal Mule, Aanchal Pandey, Shivani Kukekar and Aachal Kadu [2022], proposed a pet sing iot based. There are two ways that you can use the feeder. First, the procedure is started by pressing the feed button connected to the Arduino. When the button is pressed, the stepper motor turns, and the ultrasonic sensor keeps track of how far away it is from the container. The stepper motor opens the food valve if the distance is more than 5 cm. An alternative method of accessing the feeder is through a web server. The online interface of the feeder is revealed by using a web browser to access the Arduino's IP that is shown on



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

the LCD. Data like sensor-container distance, stepper position, and pet feeding options are displayed on this screen. When the distance is greater than 5 cm, clicking the provided link causes the stepper motor to open the food valve. The ultrasonic sensor measures the amount of food in the container. The stepper motor follows preset protocols to close the food valve when the sensor-container distance falls below 5 cm. [20]

In 2023, J. Kulaikar, D. Kurade, A. Sawant, P. Sthawarmath, and A. Chaurasia(2023) proposed a system that uses the Internet of Things (IoT). Automatic pet feeders are the solution for those who love to keep pets in the house, but because of busy schedules, they may not be able to provide food to their beloved pets at a fixed time. This prototype can be used to release food by inputting the chosen. time. Water will also be dispensed into the container through the water pump in which it will be stored. A speaker is also attached to the robot, and a voice is added to make the pet feel that the owner is communicating, which can comfort them. In addition, a camera is also fixed in the robot, which helps in monitoring the pet. Thus, the proposed prototype is expected to have immense allocation to serve the said purpose in present society, because of which pet owners can go anywhere without worrying about their pet. The project is being made so that it can be easily affordable to everyone. [21]

The Arduino Atmega 2560 serves as the primary controller in the automatic pet feeder that Tatiya Seehatrakul and Martin Viray Ocampo



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

proposed in 2023. This controller is in charge of controlling a number of sensors, processing images from cameras, presenting information on an LCD screen, and running software on electronic devices. The programming languages C and C++ are used by the system to communicate with each component, ensuring smooth functionality and integration. [22]

In 2022, P. Birha, R. Ingle, S. Tajne, P. Mule, A. Pandey, S. Kukekar, A. Kadu conducted an IoT based Pet feeder. They created a simple pet feeder using Arduino Uno as their microcontroller, they also used an Ultrasonic Sensor to detect the distance between the container, then unlike other projects using a servo motor, they integrated a stepper motor to perform the equal rotations and using a Wifi module to connect the hardware to the mobile application they created for the pet owner. Their hardware dispenses dry pet food and water to the pet. The mobile application serves as a remote controller for the pet owners to feed their pets whenever they want. [23]

DEFINITION OF TERMS

The following terms were used and defined in the study.

Operational Terms

Serial Monitor - is a tool within the Arduino Integrated Development Environment (IDE) that allows for communication between the Arduino board and a computer. It can perform a variety of actions like display, data input, communication and debugging.



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

Pet - refers to an animal that is domesticated and kept primarily for companionship or enjoyment rather than for work or utility. Pets are often considered members of the family and are cared for by their owners, providing emotional support, companionship, and entertainment. Common types of pets include dogs, cats, birds, fish, and small mammals.

Technical Terms

Arduino IDE - or Integrated Development Environment, is a software application used for programming Arduino microcontrollers. It provides a user-friendly interface for writing, compiling, and uploading code to Arduino boards.

Buzzer - An audio signaling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical. The main function of this is to convert the signal from audio to sound.

C++ Language - is a programming language used for developing software applications, system software, device drivers, client-server applications, and embedded firmware.

ESP8266 NodeMCU Module - enables wifi connectivity and microcontroller feature.

Expansion Board - These boards are designed to extend the capabilities of the



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

main board without requiring extensive wiring or complex setups.

IoT - or Internet of Things refers to the network of interconnected devices embedded with sensors, software and other technologies.

Servo motor - rotates to a specific angle based on a control signal using a controller.

Ultrasonic Sensor - measures the distance of an object using soundwaves

Chapter 3 METHODOLOGY

REQUIREMENTS ANALYSIS

To design an effective solution for PawPal, it's essential to understand the current system and procedures used by pet owners in Saint Joseph Village 10 Phase 1. The major beneficiary in this study are the pet owners and their pets residing in Saint Joseph Village 10 who will make use of the proposed automated system using IoT. Pet care and feeding is the main target the study is trying to cater to, this includes ensuring that the pets receive the appropriate portion of food at scheduled time to maintain their health and well-being, as well as ensuring that underfeeding or overfeeding is avoided at all costs. The activity also includes the pets monitoring in terms of diet, meal schedules, food container levels, dispensing notification, and alarms for low container level. Feeding typically takes place in designated areas within their homes, such as



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

kitchens or pet feeding areas. Some pet owners may also feed their pets outdoors in yards or gardens. Feeding times are generally scheduled at specific intervals throughout the day, such as in the morning, at noon, and in the evening. The timing of these activities is crucial for maintaining pets' health, and regular monitoring and adjustments are necessary to ensure the feeding schedule meets the pets' needs.

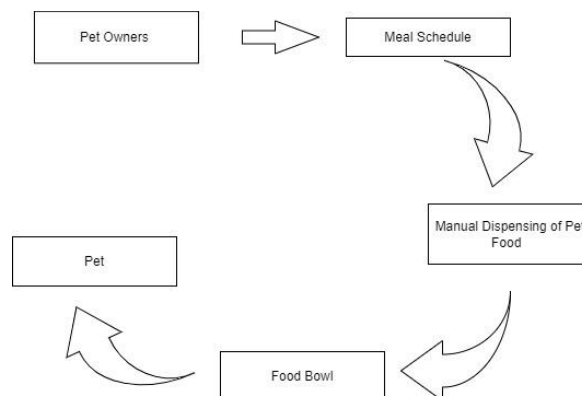


Figure 1. Current feeding methods of pet owners in Saint Joseph Village 10

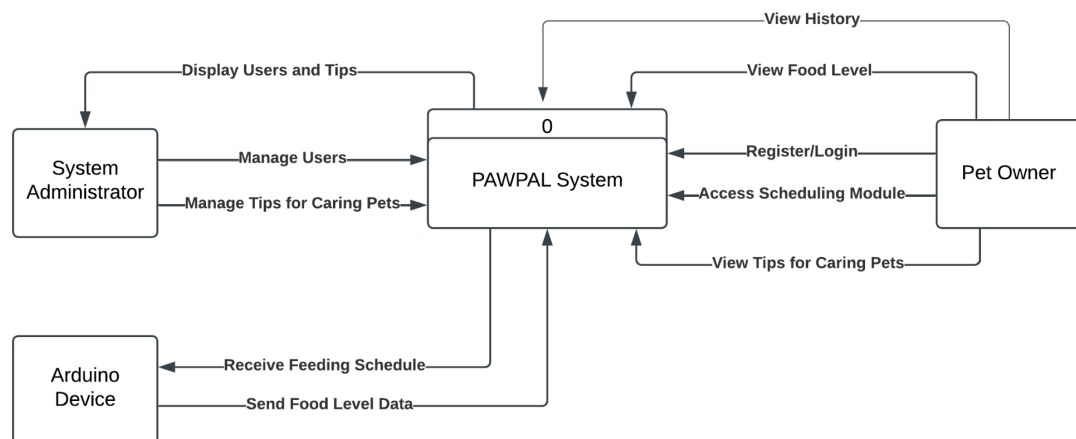
In Figure 1, currently, pet owners in Saint Joseph Village 10 primarily rely on manual feeding processes. This involves preparing food portions by hand and placing them in feeding bowls at their preferred times. Owners observe their pets during feeding to ensure they eat properly and sometimes record the details of each feeding session. Adjustments to food quantities and feeding



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

times are made based on these observations or following veterinary advice. However, this process is time-consuming and requires the constant presence of the owner or caretaker.

There are several issues with the current manual feeding system. Firstly, feeding times can be inconsistent due to the varying schedules of pet owners daily activities, which can lead to potential health issues for pets. Secondly, manual portion control is often inaccurate, resulting in overfeeding or underfeeding. Thirdly, monitoring feeding habits manually is inconvenient and prone to errors. Additionally, the lack of automation means the feeding process requires the physical presence of the owner or caretaker, which is challenging when they are away. Lastly, without automated systems, it is difficult to correlate feeding habits with health issues effectively. The following shows the flowchart for PawPal Automatic Pet Feeder.





PAMANTASAN NG LUNGSOD NG MUNTINLUPA

Figure 2. Context Diagram of the proposed system.

Figure 2 shows how "PawPal: An Android-Based Smart Pet Feeder" works for pet owners in Saint Joseph Village 10. The system has two main users: Pet Owners and Admin. Pet owners has the capability of using the functionalities of the application, such as viewing the creating an account to use the start the application, viewing the current food level inside the container, having the right to access the scheduling meal interface where they can add, edit or delete the set feeding time, and lastly is viewing the pet care tips where useful pet care tips will be given. The system administrator on the other hand is responsible for the overall viewing and manipulating of users along with their information. They have the right to display and edit tips for the pet care interface, manage user information, and manage the overall tips for caring pets. The application is connected to the internet of things to perform expected functions like receiving the feeding schedules made by the owner and sending food level data to be displayed in the application, visible to the pet owner upon accessing the interface.



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

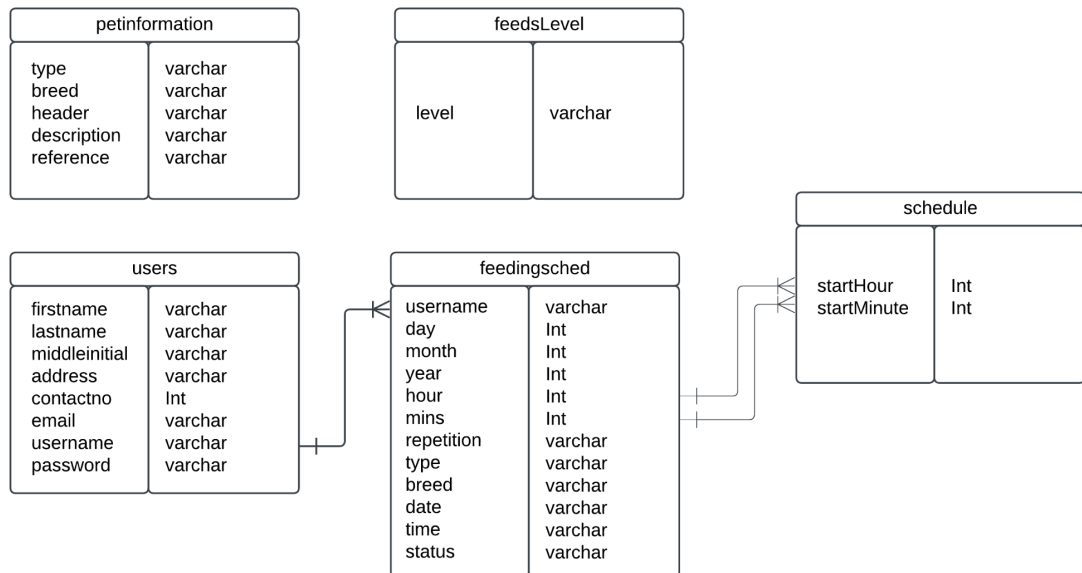


Figure 3. Entity-Relationship Diagram of the proposed system

Figure 3 illustrates the Entity-Relationship Diagram of the proposed system. The diagram demonstrates the relationship between the entities like the petinformation, feedsLevel, users, feedingsched and schedule. petinformation stores the type of pet, breed of the pet, header, description and reference which is identified by the user. feedsLevel includes an attribute, level, responsible for defining the food level inside the container (low, medium, high). The users table stores user information such as their firstname, lastname, middleinitial, address, contactno, email, username, and password having username as the identifier for the user. The feedingsched table on the other hand, stores information like the day, month, year, hour, mins, and repetition of the given meal schedules. Lastly,



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

the schedule table, which stores startHour and startMinute for the general timing of the pet feeder.

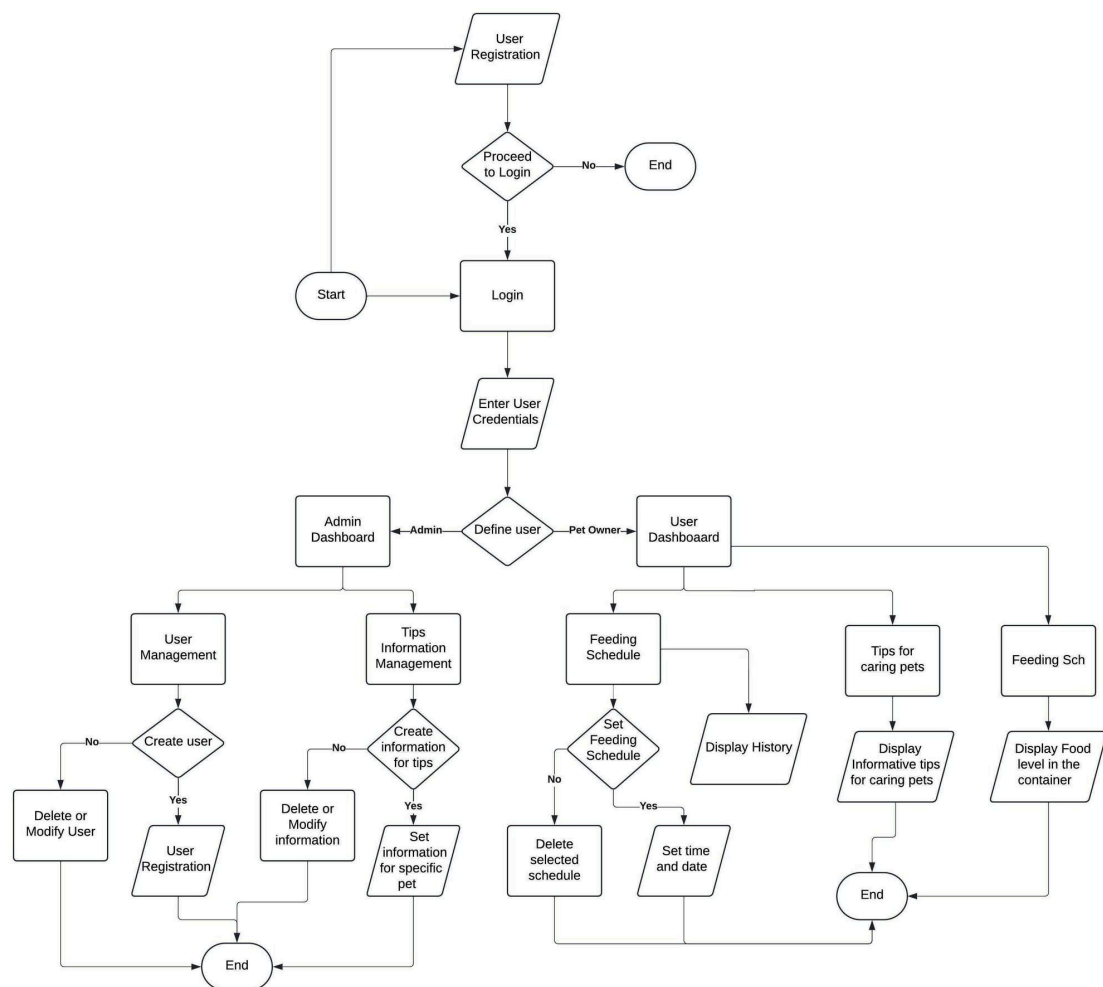


Figure 4. Flow Chart of the proposed system.

Figure 4 shows the flowchart of the proposed system. The system starts by requiring the users to create an account if it is their first time using the application, then after registration they will be redirected to the login interface. If



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

the user did not push through registration, the whole process will terminate immediately. After successfully logging in, the user will be defined by selecting either Admin or pet owner options. If Admin Dashboard, the user will be redirected to the admin dashboard having two major options namely User Management and Tips Information Management. Under User management, admin has the ability to create, modify and register new users. Under Tips Information Management, admins can add pet information tips of pet breeds available. The admin can also modify the tips added and set specific information about the pet breed available, the process will then terminate after the action is completed. If the User Dashboard is selected, there are two options available, namely Feeding Schedule and Tips for Caring Pets. Under the feeding schedule, the pet owner can set the feeding schedule and after setting the time and date, the process will end. The pet owner can also remove the existing meal schedule by simply selecting the schedule. Under the tips for caring pets, pet owners can see the informative pet tips then after selecting the breed and viewing, the process is done. Lastly, the food monitoring, the pet owner can see the current status of the food container level then after the action is done, the process will end.



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

REQUIREMENT DOCUMENTATION

The purpose of this project is to create a system that can automatically dispense food at scheduled times or based on the pet's needs to ensure consistent feeding and proper portion control. This project includes the design, development, and testing of an automatic pet food dispenser system. The system will include a food storage container, a dispensing mechanism, a schedule management system, and necessary sensors and actuators controlled by a NodeMCU.

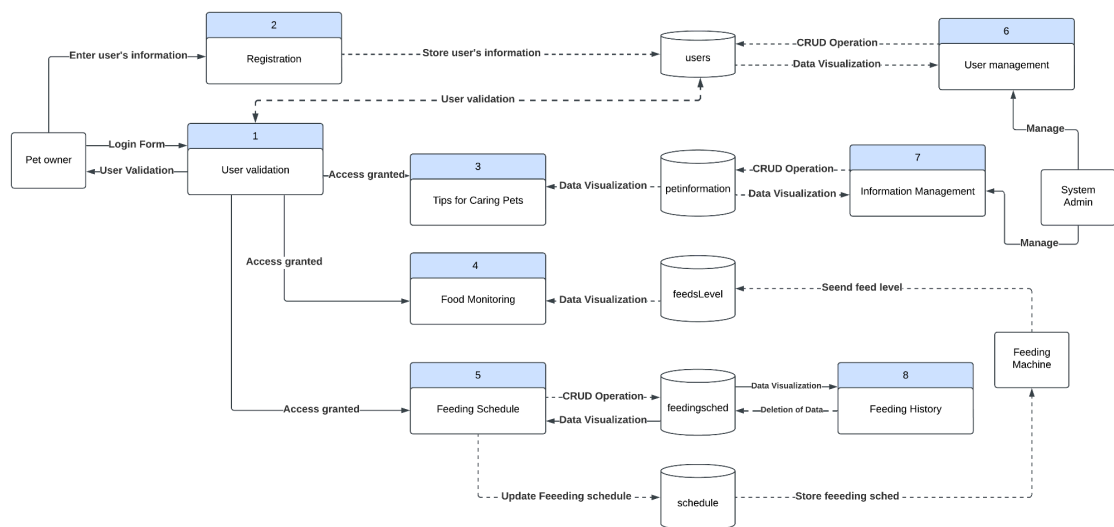


Figure 5. Data Flow Diagram of the proposed project.

Figure 5 is the Data Flow Diagram that shows how "PawPal: An Android-Based Smart Pet Feeder for Pet Owners in Saint Joseph Village 10" works. PawPal is the main system that helps pet owners feed their pets. There are two main groups using PawPal: the pet owners and the admin. Pet owners



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

give information about their pets and control it through the system. They also get automatic feeding for their pets. Admins update details about pets and keep the system working well. Inside PawPal, there are different parts called subsystems. The "Managing Pet" part takes in pet information from owners and updates it. "Feeding" part gets feeding schedules and gives out food. It also checks how much food is given. "User Notification" part tells owners about feeding and food levels, sending messages too. "Food Container Level" part manages food and tells the system about levels. It also updates owners about food levels. People interact with PawPal differently. Owners tell PawPal about pets and get automatic feeding. Admins manage the system and update pet details. PawPal works with the subsystems to handle pet information, feeding, notifications, and food levels. Subsystems let owners know about food and feeding schedules. The entire system is designed to ensure that the pet gets fed according to the schedule set by the pet owner, thereby maintaining regular feeding times without the need for the owner's constant presence.



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

DESIGN OF SOFTWARE, SYSTEMS, PRODUCT AND/OR PROCEDURE

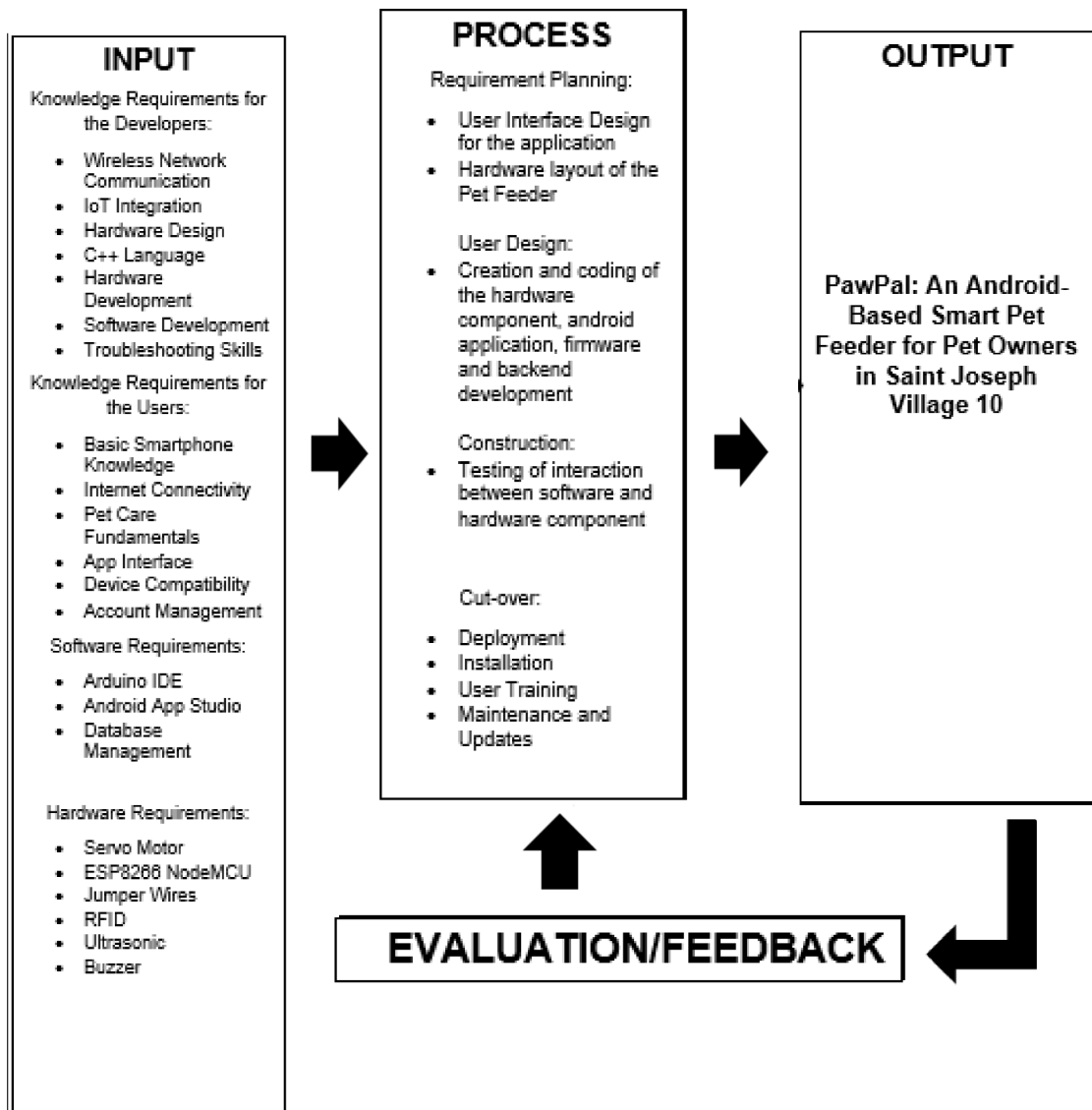


Figure 6. Conceptual Framework of the proposed system.

PawPal is an innovative smart pet feeder designed specifically for pet owners in Saint Joseph Village 10, changing the way pets are fed. This



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user-friendly, Android-based device enhances the feeding process, making it easier and more convenient for busy pet owners.

The process starts with careful planning, which includes crafting an intuitive user interface for the mobile app and designing an ergonomic layout for the feeder itself. During the development phase, it focuses on building and coding the hardware components, the Android application, firmware, and backend systems using reliable technologies such as C++, Arduino IDE, and Android Studio. This ensures that every aspect of PawPal is optimized for performance and reliability. Once the development is complete, next is to conduct thorough testing to ensure seamless integration between the software and hardware. This thorough testing phase is crucial for identifying and resolving any potential issues, guaranteeing that PawPal operates appropriately. After successful testing, is time to move on to deployment, which includes installation and comprehensive training for users. Believing that empowering pet owners with knowledge about how to use PawPal effectively is key to enhancing their experience. To ensure PawPal continues to operate at peak performance, there are offer like ongoing maintenance and regular updates. This commitment to support means that users can rely on PawPal for their pet feeding needs without worry.

By leveraging cutting-edge IoT and wireless technology, PawPal seamlessly integrates into the daily lives of pet owners, transforming pet feeding routines



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into a more convenient and stress-free experience. With PawPal, pet owners can enjoy peace of mind knowing their furry friends are well-fed, even when they're not at home.

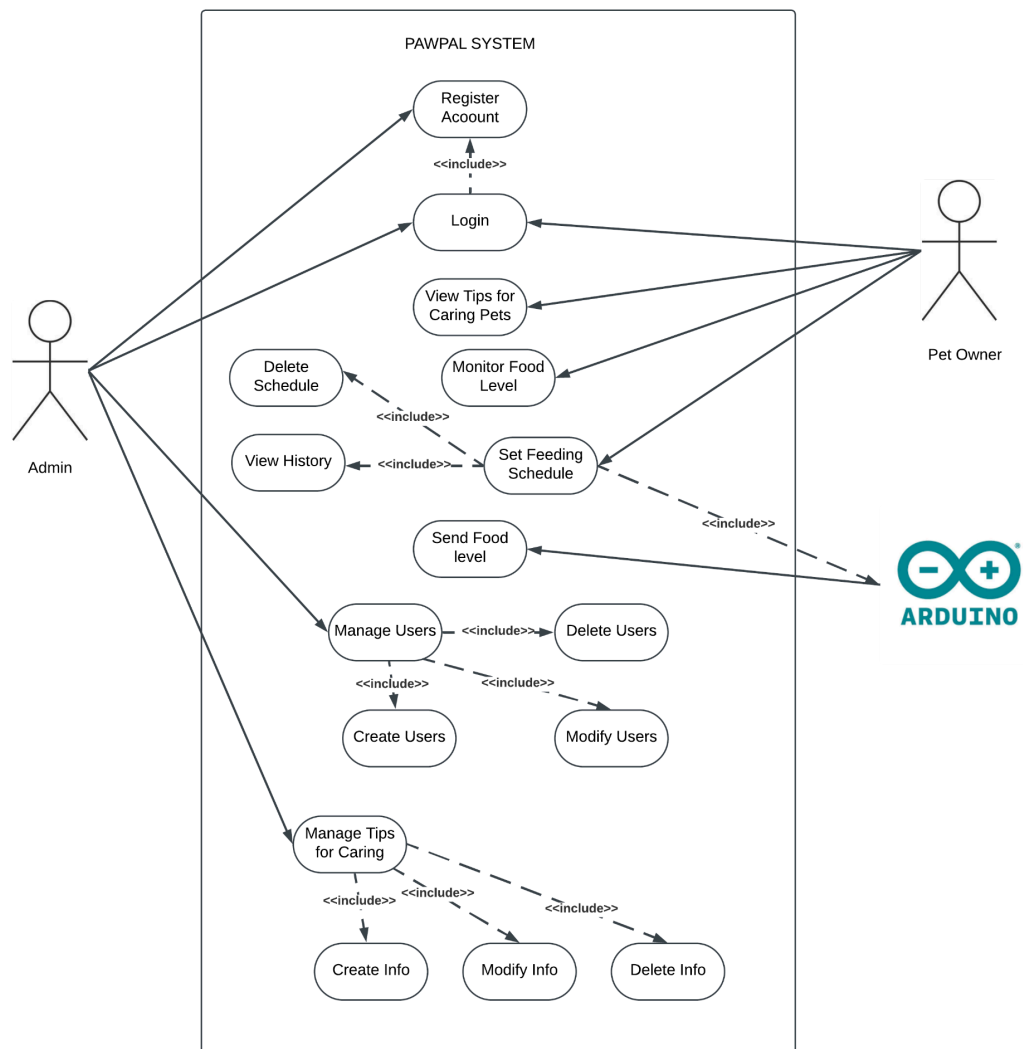


Figure 7. Use case diagram of the proposed Android-Based Smart Pet Feeder



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Figure 7 illustrates the interactions within "PawPal: An Android-Based Smart Pet Feeder," highlighting three main users: the Pet Owner, the Admin, and the Pet. The Pet Owner can select the pet breed, schedule meals, check food levels, monitor feeding, and receive notifications through the application. The Admin is responsible for updating pet information and maintaining the system to ensure it functions correctly. This diagram captures the essential features and user roles within the PawPal system, showcasing how the application helps in managing pet feeding routines efficiently and ensuring the pet is well-fed and cared for.

DESCRIPTION OF THE PROTOTYPE

Schematic diagram

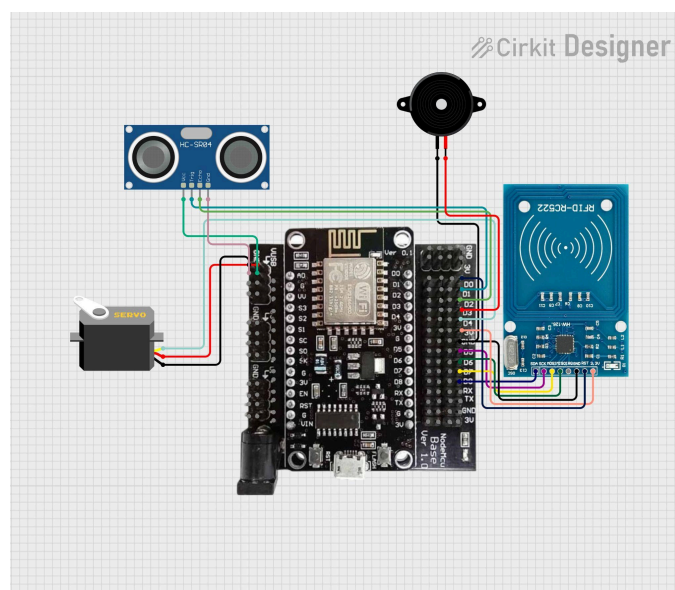


Figure 8. Schematic diagram of the proposed Android-Based Smart Pet Feeder



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Buzzer - an audio signaling device which may be mechanical, electromechanical, or piezoelectric.

ESP8266 Node MCU - used for the WiFi and bluetooth connectivity fit for the remote controlling and monitoring.

Expansion Board - facilitates seamless connectivity with support power. It simplifies pin connections, enabling effortless interfacing with external devices.

RFID - used for manual override of the pet feeder.

Servo Motor - for the dispensing of the feeder. Rotates on a specified angle to release food.

Ultrasonic Sensor - measures the distance between the food container and the pet feed to identify food container level.

DEVELOPMENT AND TESTING

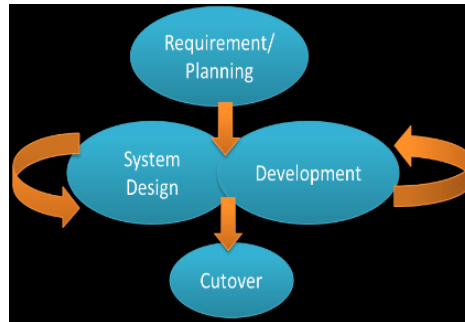
Development Procedure

PROJECT DEVELOPMENT:

Applying the RAD methodology to the development of an Automatic Pet Feeder using NodeMCU enables a quick turnaround from concept to functional prototype. With its iterative nature and emphasis on user involvement, RAD is well-suited for projects like this where rapid prototyping and continuous feedback are crucial.



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Source: ResearchGate

Figure 9. Rapid Application Development (RAD) model of Android-Based Smart Pet Feeder

Rapid Application Development (RAD), with its focus on rapid development and adaptability, is a great fit for creating an Automatic Pet Feeder. They can quickly create several feeder prototypes using RAD, which enables to evaluate their functionality through pet and owner testing in real-world settings. By using an iterative process, they may make necessary design adjustments to the feeder based on insightful input, resulting in a feeder that perfectly suits the requirements and preferences of pets and their carers. Moreover, RAD speeds up the development process, allowing to release the feeder onto the market sooner and offer pet owners a practical option as soon as possible. Furthermore, RAD facilitates the prompt detection and handling of any problems, lowering the possibility of major setbacks as the project moves forward. By working closely together and maintaining an adaptable mentality, RAD enables to develop an exceptional Automatic Pet Feeder that improves the quality of life for both pet owners and their animals.



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Phase 1: Requirement Planning

In order to gather particular demands across several important areas during the first phase of building an Automatic Pet Feeder using the Rapid Application Development (RAD) approach, The researchers spoke with pet owners in-depth and through surveys and interviews. Among these are the following: automatic time scheduling to accommodate pets' feeding schedules; up-to-date information on feeding schedules; Android compatibility for mobile device monitoring and remote control; and food container level that adjusts automatically to ensure precise and customized nutrition. By including stakeholders in these conversations and soliciting feedback on these fundamental components, They provide the groundwork for a feeder that effectively satisfies the diverse needs of dogs and their foster caregivers. This stage of the RAD process ensures that the design decisions closely align with user expectations by serving as a framework for subsequent development phases.

Phase 2: User Design

Prototyping and iterative development are the next steps in the RAD methodology's development of the Automatic Pet Feeder. First, researchers build working prototypes that include important functions like portion control, automatic feeding, and mobile device compatibility. These prototypes provide concrete depictions of the design and operation of the feeder. These prototypes



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are put through real-world testing with pets and their owners, which gives critical feedback on usability and the overall user experience. They make changes to the prototypes in response to these comments, fixing any problems or suggesting areas for development. We keep lines of communication open with pet owners during this process to make sure the feeder adapts to their changing needs and preferences.. To ensure that the feeder meets the standards for dependability, safety, and reliability, technical validation is performed on a regular basis. By embracing rapid prototyping and iterative development, we can fast iterate on the Automatic Pet Feeder design. Based on user feedback and technical validation, we may develop the design to create a solution that eventually satisfies the demands of dogs and their owners while providing an enhanced user experience.

Phase 3: Construction

In Phase 3, we improve the Automatic Pet Feeder by utilizing the insights gathered from the Phase 2 testing. Using Rapid Application Development (RAD), we quickly apply changes based on feedback from pet owners. Android Studio, the Arduino IDE, the C++ programming language, and the NodeMCU are a few of the software tools we use to enhance the design and functionality of the feeder. We test multiple versions of the feeder to see which one suits pets and their owners the best. Through this iterative approach, we may enhance the feeder's features—like portion control, scheduling, and alerting systems—to



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better meet the needs of pets and caregivers. By continuously improving the feeder in response to feedback from the real world, we ensure that it provides pets and their owners with a seamless and dependable feeding experience. Our collaborative skills and RAD's flexibility enable us to create a superior Automatic Pet Feeder that enhances the well-being of pet owners and their furry companions.

Phase 4: Cut-over

We ensure that everything is ready for mass manufacturing. We finish the design and set up our manufacturing procedure so that we can make feeders fast without compromising on quality. Next, we present the feeder to the public and let them know where it is and what its features are. We make sure pet owners are aware of the potential for life-simulation. We assist customers with the setup and operation of the feeder. We immediately address any issues that arise. We keep gathering feedback from users to figure out how we can make the feeder even better. We aim to continuously improve it by listening to pet owners' feedback. We ensure that feeds containing software, Android Studio C++, or the Arduino IDE, for example are updated. They preserve its functionality and incorporate new features in this way. We routinely check the feeder's functionality and safety, and we respond fast to any problems that may occur. By taking these precautions, we guarantee that our Automatic Pet Feeder



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will always be a top-notch item that enhances the lives of pet owners and their animals.

Testing Procedure

To ensure that the tools are working properly, hardware and software testing is conducted as follows:

Table1. Testing procedures of the proposed project.

Testing phase	Procedures	Results
1. Testing of components	A. Servo Motor - opening and closing on a certain angle and the interval of delay to monitor how fast it and how much it will dispense. B. Ultrasonic Sensor - detecting the distance between the container and the pet food to detect current level. C. NodeMCU Module checks if the module successfully connects the application to the internet. D. Buzzer - sends an alarm notification to the owner if the container is running low. E. RFID - for manual operation in connection with the servo motor using the sensor and tags.	The components harmonize together and perform the expected functions.
2. Feature Testing	Test all the functions and features of the automatic pet feeder and check if the pet feeder automatically dispenses at the scheduled time using the mobile application. Ensure that the feeder responds to the developed application using Android Studio and see if it connects successfully to the internet.	The pet feeder automatically dispensed the food on the scheduled time along with the other features. The Android Studio app is working properly and connects successfully to the internet.



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3. User-Testing	Provide a real time environment testing for the users. Test if the interface is comprehensive and user-friendly and see if the project performs the specified features.	The pet owners find the pet feeder useful and convenient. Serving them the features that will help with the remote feeding of their pets with the automated features.
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Table 1 is the evaluation instrument based on the ISO/IEC standard. The target respondents of this study will be 20-30 pet owners in Saint Joseph Village 10 Phase 1 who are the main proponents of the pet feeder and 10 Information Technology (I.T) experts. Survey and interview will be conducted, surveys are a form of data gathering techniques consisting of a series of questions. Interview on the other hand, is a face to face conversation between respondents and the researchers. Both are ideal for the pet owners in Saint Joseph Village 10 to assess their pets eating habits and demographics.

Table 2. Rating scale used in the project instrument.

Test	Description
Functional Suitability	<i>Refers to whether a software application or system fulfills the functional requirements and satisfies user needs.</i>
Performance Efficiency	<i>Refers to how effectively and efficiently the system executes its intended tasks</i>
Compatibility	<i>Refers to the capability of the system to work seamlessly with other systems and software.</i>
Usability	<i>Refers to the simplicity and how usable the system is for its target users.</i>
Reliability	<i>Refers to the system's reliability and ability to operate consistently without errors.</i>
Security	<i>Refers to the system's measures to safeguard</i>



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	<i>access and protect sensitive information.</i>
Maintainability	<i>Refers to the ease of maintaining and updating the system over time.</i>
Portability	<i>Refers to the system's compatibility with various platforms and environments.</i>

Table 2 shows the areas to be evaluated using the ISO/IEC 25010:2011 Software Evaluation Tool. It is curated to evaluate the system's functionality, performance efficiency, compatibility, usability, reliability, security, maintainability, Nodeand portability to ensure that it meets the standards.

Table 3. Rating scale of the proposed project.

Scale	Rating	Interpretation
5	4.50 - 5.00	Excellent (E)
4	3.50 - 4.49	Very Good (VG)
3	2.50 - 3.49	Good (G)
2	1.50 - 2.49	Fair (F)
1	1.00 - 1.49	Poor (P)

Table 3 shows the likert scale used in the evaluation. Having 5 as its highest and 1 being the lowest. 5 is equivalent to an excellent interpretation whereas 1 is poor as its interpretation.



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IMPLEMENTATION PLAN

The PawPal: An Android Based Automatic Pet Feeder will start by project planning and by conducting project analysis. After doing so, the group will then proceed to development of hardware of the pet feeder, including all the necessary parts that will hold all the necessary components later on. The android application development comes next after the completion of the hardware component, here the researchers will make use of the Android Studio App to develop a fully functional application which will be integrated to the software. This will be then integrated all together to complete the automatic pet feeder and test the functionality and check if the pet feeders meet the features it was made to do. The group will ask a permit to the President of the Home Owner's Association of Phase 1 for the implementation of the project and to conduct user testing in the said area. The timetable below shows the activity schedules with its corresponding duration.

Table 4. Timetable of the activities for the proposed project.

ACTIVITIES	PERSON INVOLVED	DURATION
Project approval & Requirement Analysis	Researchers / Adviser	1 week
Hardware Development	Researchers	2 weeks
Android App Development	Researchers	2 weeks



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Phase 1 HOA Permit for Testing	Phase 1 HOA President / Researchers	1 day
Integration and Testing	Researchers/ Target Users	5 days
Deployment & Launch	Researchers / Target Users	3 days

Table 4 shows the timetable for the creation and deployment of the system. The table shows the activities to be done, people involved and the duration period of each activity. It provides a detailed description of the implementation plan which will serve as a guide for the users and the researchers.

CHAPTER 4

RESULTS AND DISCUSSION

PROJECT DESCRIPTION

Pawpal is an Android Based application designed to cater busy pet owners of Saint Joseph Village 10. The system aims to improve the pet's feeding through an innovative integration of IoT to provide automated feeding and monitoring.

The system is a functional automated pet feeder with an application to manipulate the feeder and schedule meals for their pets. The system is created with a registration form in the beginning for the user to create an account before using the application. Then there are three major functionalities, the feeding



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schedule; for the setting, viewing and manipulating of the scheduled meals, food level status and pet care for providing useful tips and reminders to the owners on how to take care of their pets in line with their specific needs. Then the food monitoring for the container updates with regards to the current food level inside the container. The system is developed to alleviate the struggles of pet owners in the dietary aspect of their pets and to provide a user-friendly android application to further explore the capabilities of the system.

PROJECT STRUCTURES

The following components of the project are presented both the system and the prototype, which is developed and constructed to be used by the pet owners in Saint Joseph Village 10.

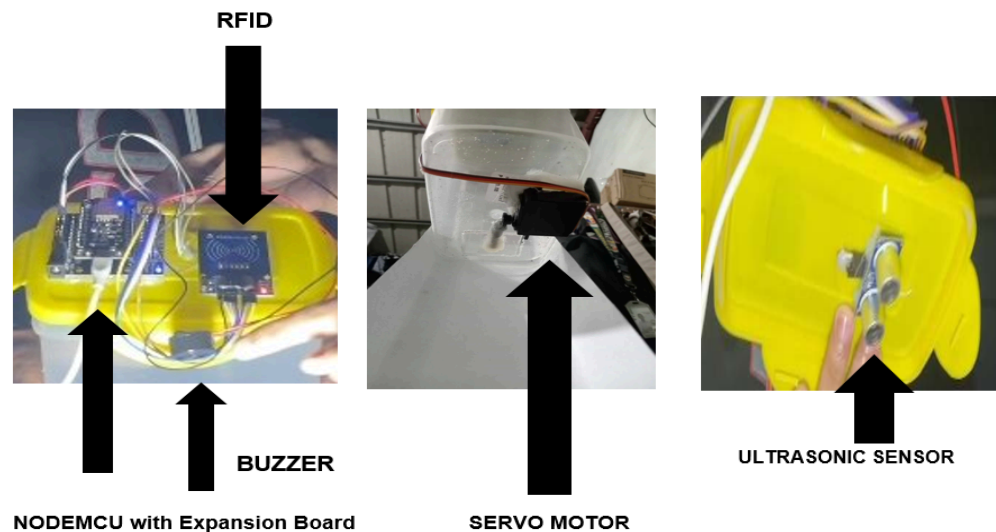


Figure 10. PAWPAL: Android-Based Smart Pet Feeder

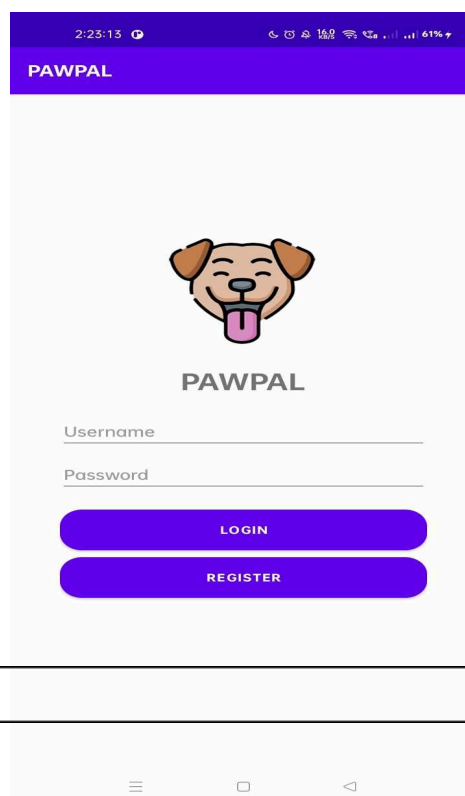
Figure 10 shows the prototype and the components we use for our pet feeder.

- NodeMCU with Expansion Board



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- This component serves as the microcontroller of the PawPal pet feeder where all the components are connected to the internet.
- RFID
- This component is used to operate the feeder manually by using the RFID tags.
- Buzzer
- This component is used to notify the owner if the food level is low or empty.
- Ultrasonic Sensor
- This component is used to sense the level of the food and send it to the buzzer to alarm the owner if the food level is low or empty.
- Servo Motor
- This component is used for food dispensing mechanisms.





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Figure 11. Login Page of Android-Based Smart Pet Feeder

The PawPal app has two main dashboards: the User Dashboard and the Admin Dashboard. The login and registration feature is the first step for both users and admins. The login screen asks for a username and password to make sure only the right people can access their accounts. New users can also sign up by creating an account, which helps more pet owners start using the app.

Figure 12. Registration Form Page of Android-Based Smart Pet Feeder



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The PAWPAL registration form is a simple way to sign up for an account. It asks for basic information like your last name, first name, middle initial, email, address, and phone number. Users also need to create a username and password and confirm the password to make sure there are no mistakes. After filling out all the fields, They can click the "REGISTER" button to complete the sign-up process. It's an easy and quick way to get started with PAWPAL.

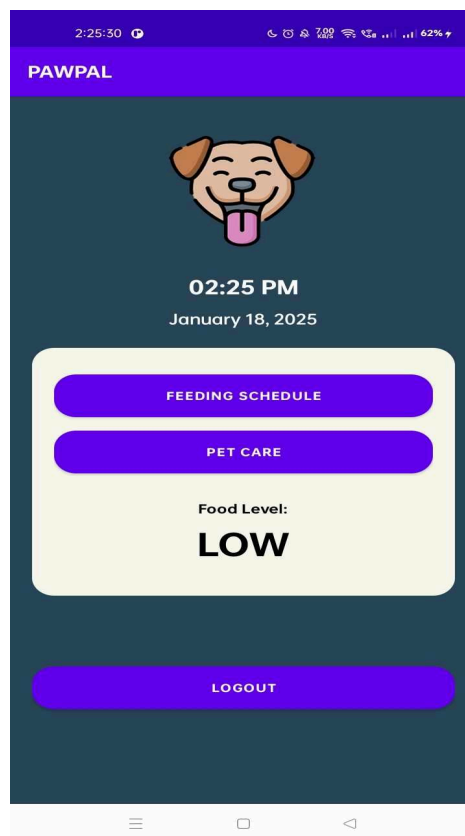


Figure 13. User Dashboard Page of Android-Based Smart Pet Feeder



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This is the user dashboard for a pet care app. It's likely the first screen users see after logging in. The interface shows the current time and date, and quick buttons for "Feeding Schedule," "Pet Care," and "Logout." There's also a "Food Level" section that shows the pet's food container is currently "Low," which is a helpful alert for the user. These features let users easily see important information about their pet's needs and access key functions without going through many menus.

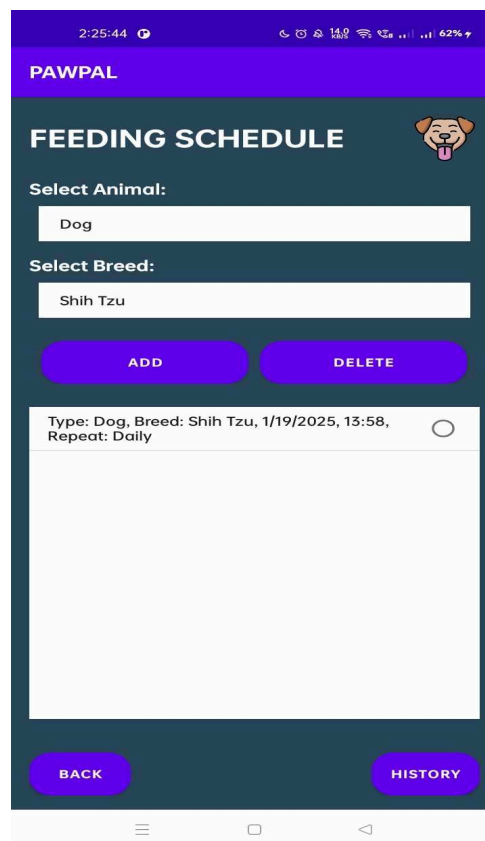


Figure 14. Feeding Schedule Page of Android-Based Smart Pet Feeder



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This is a feeding schedule which is used to create feeding schedules for pets. The app allows users to select an animal and its breed, as shown in the image. They can also add or delete additional feeding schedules using the buttons "ADD" and "DELETE" on the screen. Lastly, users can go "BACK" to the previous screen or go to the "HISTORY" of the schedules created.

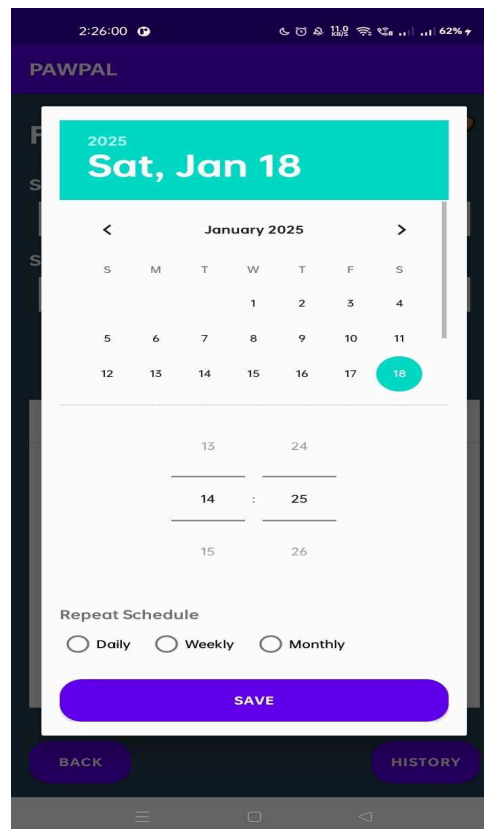


Figure 15. Adding Schedule Page of Android-Based Smart Pet Feeder

This is the add function in the feeding schedule like setting up a date and time through a calendar section. The time format is in military time format, as this is the format compatible with our database. There is a section in which the



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user can choose the frequency of feeding schedule. They have chosen to repeat the schedule on a weekly basis. The user has the option to save the schedule or go back to the previous screen.



Figure 16. History Page of Android-Based Smart Pet Feeder

The history page shows the records of the feeding schedule that are already completed. It shows information like the type of dog , breed and the date and time and also a status. Then the clear history button for deleting the records



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of the feeding history . This enables the user to view the logs for the feeding feature of the system.

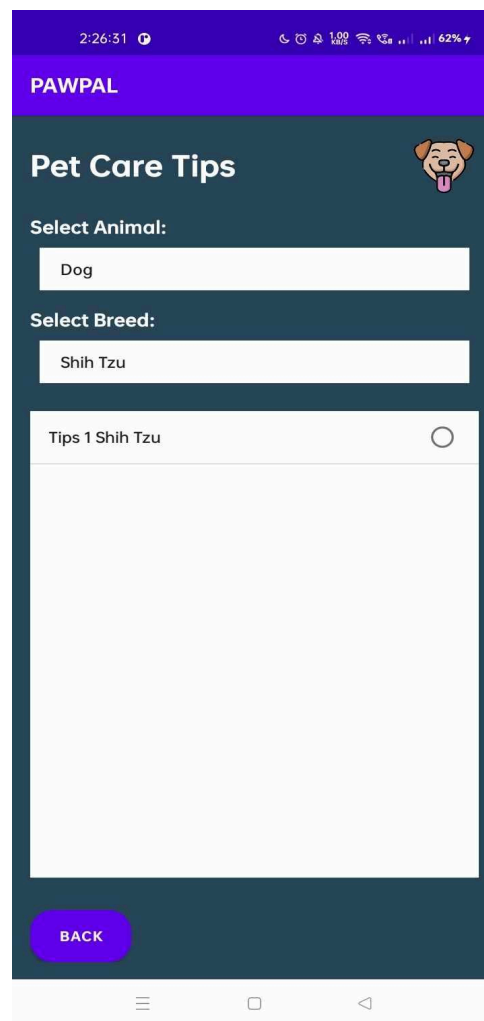


Figure 17. Pet Care Tips Page of Android-Based Smart Pet Feeder

This is the pet care tips page, this page is used to give the owners useful tips especially if the user is a first time pet owner. The user can select a dog or



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cat as the type of animal and there are 8 different available types of breeds. The app then displays a section of tips that helps the owner on how to manage and take care of their pets .

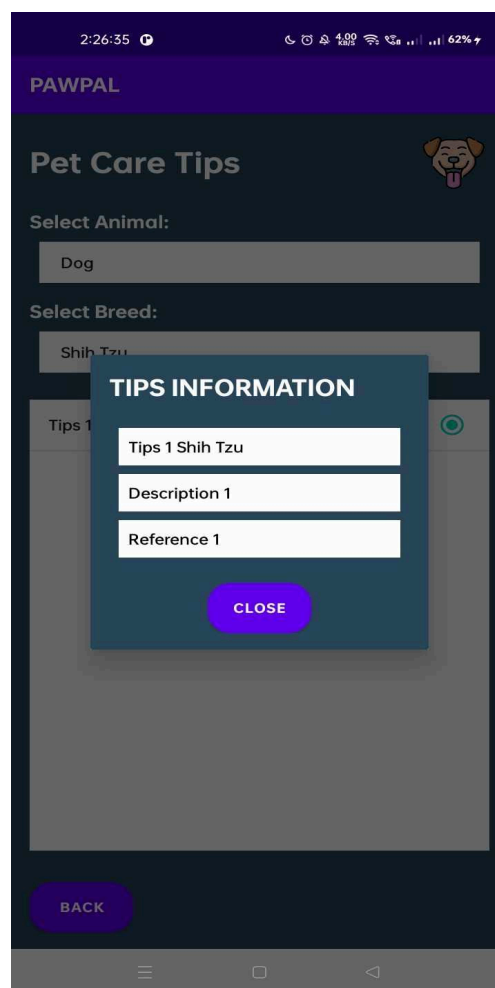


Figure 18. Pet Care Tips Information Page of Android-Based Smart Pet Feeder

This is pet care tips information. It is an interface of the app that gives pet care tips. They are presented with a pop-up window that displays 3 three



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text-fields. The admin can fill these fields to add information about dog and cats care tips. This app seems to be designed to allow users to learn pet care tips for different breeds.

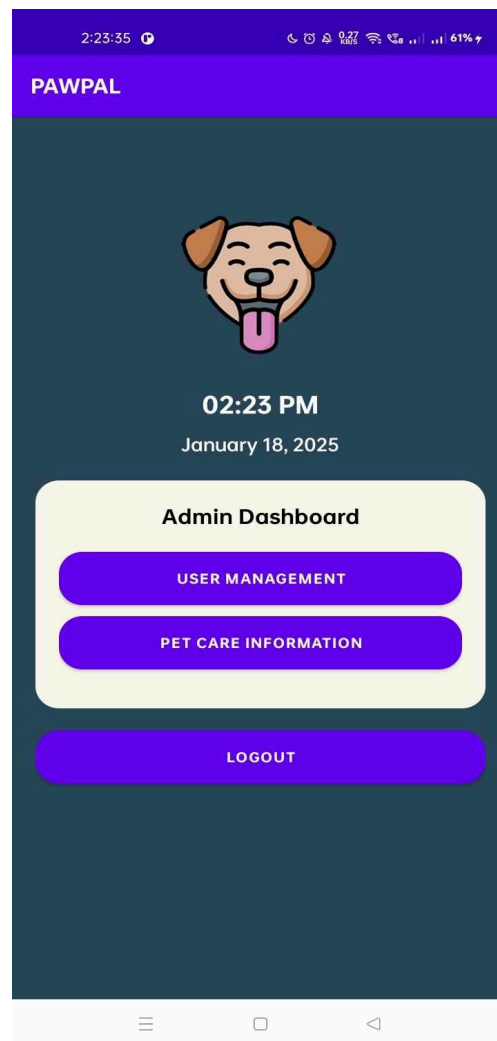


Figure 19. Admin Dashboard Page of Android-Based Smart Pet Feeder

This is the Admin Dashboard page. It has the same interface like the user dashboard but the differences are the buttons and its functionality. There are two



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main buttons in the admin dashboard, the user management and the pet care information and a logout button to exit the dashboard.

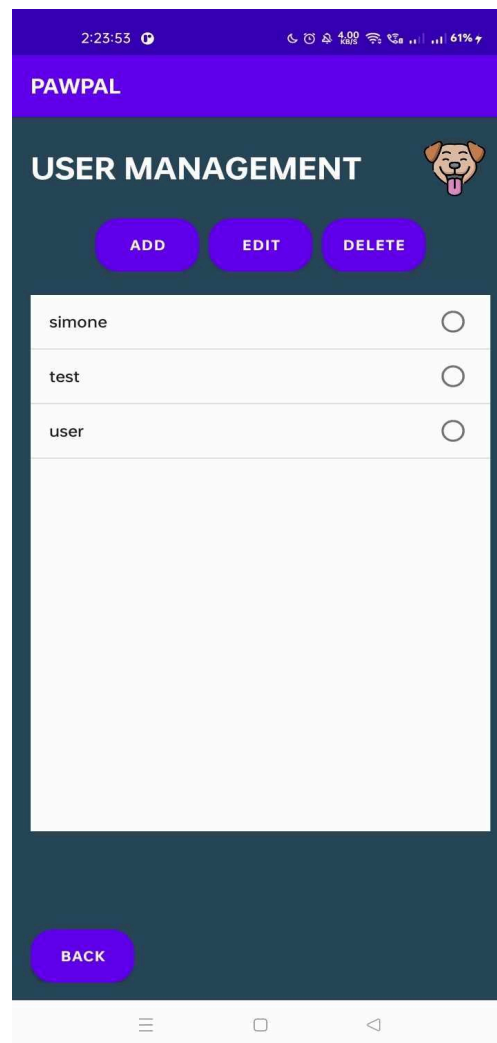


Figure 20.. User Management Page of Android-Based Smart Pet Feeder

This is the user management page. This page is part of the admin dashboard and is used to manage the accounts of the user. This page also has



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a list of accounts for users and it can also add, edit and delete the accounts of the current user.

The screenshot shows a mobile application interface for 'PAWPAL'. The top status bar shows the time as 2:24:15 and a battery level of 61%. The app's header is purple with the text 'PAWPAL'. Below the header, the title 'USER MANAGEMENT' is displayed in a dark blue bar, accompanied by a small dog icon. Underneath the title, there are three buttons: 'ADD', 'EDIT', and 'DELETE'. The 'ADD' button is selected, and a white modal form is displayed over the app. The form contains the following fields: 'Last Name', 'First Name', 'Middle Initial', 'Email', 'Address', 'Contact No.', 'Username', and 'Password'. At the bottom of the form is a purple button labeled 'ADD USER'. A 'BACK' button is also visible at the bottom left of the screen.

Figure 21. Add User Page of Android-Based Smart Pet Feeder

This is the add user window in the user management interface. Admin can add new users by creating their accounts, same procedure like the registration form page. It asks for basic information like your last name, first



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name, middle initial, email, address, and phone number. They also need to create a username and password and confirm the password to make sure there are no mistakes before adding it using the add user button.

The screenshot shows the PAWPAL app interface. At the top, the status bar displays the time 2:24:20 and battery level 61%. The app header is purple with the text 'PAWPAL'. Below the header, the title 'USER MANAGEMENT' is displayed in white, accompanied by a small dog icon. A modal form is open for editing a user. The form contains the following fields: 'Mendez' (Name), 'Simone' (Middle Initial), 'B' (Last Name), 'simonemendez02@gmail.com' (Email), 'San Pedro Laguna' (Address), '09876356215' (Phone Number), 'simone' (Username), and a password field with six dots. A purple 'UPDATE USER' button is at the bottom of the form. A 'BACK' button is located at the bottom left of the screen.

Figure 22. Edit User Page of Android-Based Smart Pet Feeder

This is the edit user page for user management. The admin can make changes to a current account. The admin needs to input information on fields that need updating and click the “Update User” to save.



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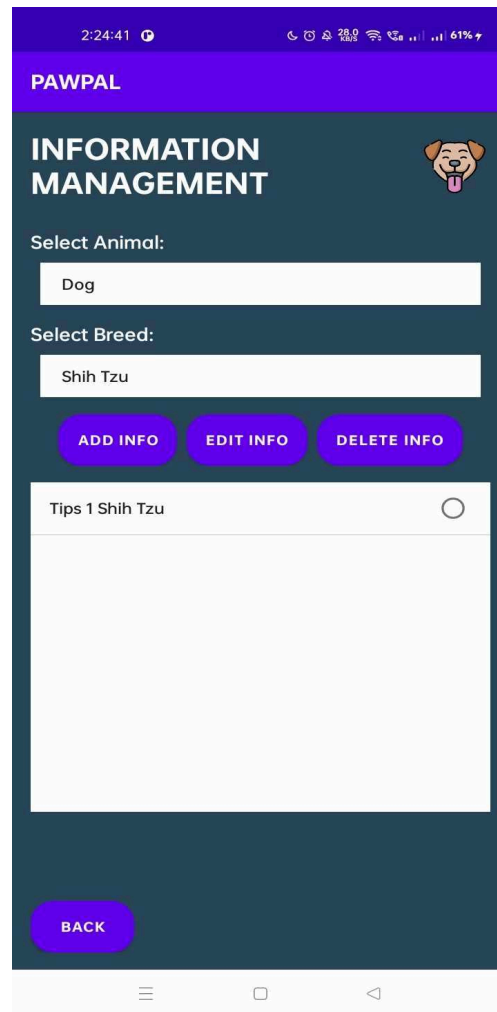


Figure 23. Pet Care Information Management Page of Android-Based Smart Pet Feeder

Aside from the user management button there is a button for pet care information. This page is used for adding , editing and deleting tips and information about pet care. It also has a panel so that admin can see the list of each tip according to different breeds.



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The screenshot shows a mobile application interface for 'PAWPAL'. The main screen is titled 'INFORMATION MANAGEMENT' and has a dark blue background. A modal window titled 'ADD CONTENT' is open in the center. The modal has a light blue header and contains the following fields: 'Animal Type:' with a dropdown menu showing 'Dog', 'Breed:' with a dropdown menu showing 'Shih Tzu', and 'Information:' with three text input fields labeled 'Enter Header', 'Enter Description', and 'Enter Reference'. At the bottom of the modal are two buttons: 'SAVE' and 'CANCEL'. The background app interface shows a 'Select Animal:' dropdown with 'Dog' selected, a 'Select Breed:' dropdown with 'Shih Tzu' selected, and a 'Tips' section. A 'BACK' button is visible at the bottom left of the app screen.

Figure 24. Pet Care Information Management Add Page of Android-Based Smart Pet Feeder

This is the panel of Adding information in pet care. The admin can select whether dogs or cats and different available breeds, then under information, the admin can put the tips , descriptions and the reference for that. The admin can click “Save” to save the information and “Cancel” to go back to the information management interface.



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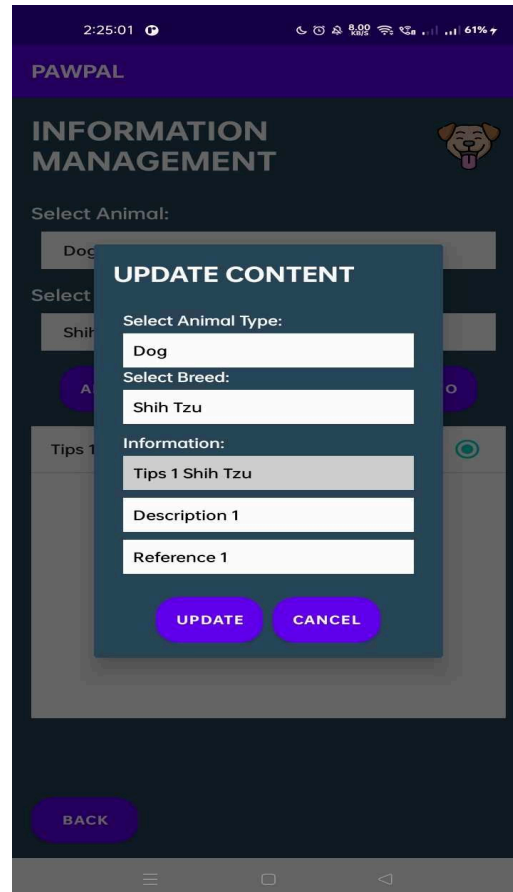


Figure 25. Pet Care Information Management Editing Page of Android-Based Smart Pet Feeder

This page is for editing and updating the information or the pet care tips. Admin can select the animal type to update information from then a breed selection tab with the corresponding information, all the data will be saved after pressing the update button and go back to the information management by pressing cancel.



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PROJECT CAPABILITIES AND LIMITATION

The following are the capabilities of Pawpal: An Android Based Automatic Pet Feeder:

- The system is capable of executing manual and remote pet feeding features.
- The system is capable of performing user registration and login.
- The system is capable of providing real time and date.
- The system is capable of setting meal schedules and their frequency.
- The system is capable of providing a real time update on the level of the food container.
- The system is capable of alerting the owner through the alarm of the buzzer.
- The system is capable of providing a pet and breed selection through a drop down menu.
- The system is capable of adding and deleting meal schedules.
- The system is capable of showing history and a clear history option for all the feeding logs.
- The system is capable of providing pet care tips and references.
- The system is capable of showing eight different pet breeds along with useful tips.
- The system is capable of performing admin functions like user



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management and updating pet care information.

- The system can support multiple feeding schedules.
- The prototype can instantly trigger an unscheduled feeding if needed by using RFID.
- The prototype has a Sensor to alert when food runs low.

Limitation

- The system is dependent on internet connection and cannot perform specified functions without a stable WiFi connection.
- The system only supports android devices with android versions 8 and up and cannot support iOS versions.
- The prototype is designed for dry foods only and dispenses uniform portions for all the breeds.

PROJECT EVALUATION

To evaluate the capabilities of the system, the system went through an actual testing and evaluation from the actual end-users and IT Professionals. Prior testing was conducted by the researchers to ensure that the system, components and functionalities are working properly before the deployment of the project. The system is evaluated by 30 target end users and 10 respected IT experts, assessing the project using an evaluation form with the following criteria: functionality, performance, compatibility, usability, reliability, security,



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maintainability, and portability of the automated pet feeder for pet owners of Saint Joseph Village 1, Phase 1.

Table 5. Testing areas

Areas to be Tested	Tests to be conducted
User Login Page	Test all the buttons in the login page and ensure that they are working. Users should be redirected to the right pages of the application and should be able to register and create an account.
Feeding Schedule Function	View the calendar and schedule meals with the corresponding date, time, day and frequency. Select, add, edit, and delete the scheduled feeding time whenever.
Pet Care Function	Show useful tips provided by the admin on how to take care of pets of the provided breeds. Dropdown buttons will show the kind of pet and the available breeds with the corresponding tips.
Logout Function	Enable users to log out and terminate the application completely. Should redirect the user back to the login/register page.



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Admin Page	View all the users of the application, manipulate information, manipulate data from different pages like the user, user information, tips for pet care.
Hardware Function	Perform functions such as manual overriding using RFID Tag, automated feeding based on the scheduled meal, and give container status update.

Table 5 shows the areas to be tested and what are the expected functions to be tested. For the login page, the function of the buttons are tested to see the interactivity of the buttons and if it redirects the user to the right pages. For the feeding schedule function, the testing emphasizes the capability of the application to provide the functions in scheduling a meal alongside the frequency of the schedule. The pet care function was tested to see if it prompts the two kinds of pets and the available breeds of each kind, it should also change the tips given the kind of pet and their breed. Next is the admin page, it evaluates the capability of the admin to perform admin functions and manipulate, retrieve and update the received data. The admin is also responsible for editing the tips shown in the pet care page of the application. The Hardware function follows, focusing mainly on the performance of the



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

hardware in terms of performing automated feeding functions using the application as a medium. Then lastly the logout function,

Table 6. Population size of target respondents.

Respondens	Sample Size
End users	30
IT Experts	10
TOTAL	40

Table 6 shows the total number of respondents for the evaluation. There are a total of thirty (30) actual users and ten (10) IT experts to evaluate the system.

The following table shows the criteria for evaluation used by the evaluators to assess the system.

Table 7. Summary of the Mean Score of the evaluation from Actual Users

Criteria	Weighted Mean	Descriptive Rating	Rank
1. Functionality	4.49	Very Good	4
2. Reliability	4.37	Very Good	5
3. Usability	4.78	Excellent	1
4. Efficiency	4.66	Excellent	2
5. Portability	4.61	Excellent	3
Overall Weighted Mean	4.58	Excellent	
Legend: Scale Numerical Rating Descriptive Rating 5 4.50 – 5.00 Excellent (E)			



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

4	3.50 – 4.49	Very Good (VG)
3	2.50 – 3.49	Good (G)
2	1.50 – 2.49	Fair (F)
1	1.00 – 1.49	Poor (P)

Table 7 shows the results of the results of the evaluation from the end users of the application. The different areas like the Functionality produced a weighted mean of “4.49” with a descriptive rating of “Very Good”. The Reliability produced a weighted mean of “4.37” with a descriptive rating of “Very Good”, Usability having a weighted mean of “4.78” with a descriptive rating of “Excellent”, Efficiency having a weighted mean of “4.66” with a descriptive rating of “Excellent”, and lastly, Portability producing a weighted mean of “4.61” with a descriptive rating of “Excellent”. The evaluation had an overall weighted mean of “4.58” deeming the whole system as “Excellent”.

The following are the results and findings gathered from the evaluation of the actual users:

1. Functionality: The respondents evaluated the hardware and software with the weighted average of “4.49” with a corresponding rating of “Very Good” because the expected functions are met and executed well.
2. Reliability: The respondents evaluated the hardware and software with a weighted average of “4.37” with a corresponding rating of “Very Good” because the system gave consistent results throughout the evaluation.



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

3. Usability: The respondents evaluated the hardware and software with a weighted average of “4.78” having a corresponding rating of “Excellent” because the application is user-friendly, easy to understand and has a very simple yet functional interface.
4. Efficiency: The respondents evaluated the hardware and software with a weighted average of “4.66” and a descriptive rating of “Excellent” because the system met the desired results and performed effectively and efficiently during the evaluation.
5. Portability: The respondents evaluated the hardware and software with a weighted average of “4.61” with a descriptive rating of “Excellent” because the hardware and software can be accessed with ease and is very convenient to use.

Table 8. Summary of the Mean Score of the Evaluation from IT Experts

Criteria	Weighted Mean	Descriptive Rating	Rank
1. Functional Suitability	4.56	Excellent	4
2. Performance Efficiency	4.6	Excellent	3
3. Compatibility	4.65	Excellent	2
4. Usability	4.45	Very Good	7
5. Reliability	4.55	Excellent	5



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

6. Security	4.52	Excellent	6																								
7. Maintainability	4.44	Very Good	8																								
8. Portability	4.7	Excellent	1																								
Overall Weighted Mean	4.55	Excellent																									
<table> <tr> <td>Legend:</td><td>Scale</td><td>Numerical Rating</td><td>Descriptive Rating</td></tr> <tr> <td></td><td>5</td><td>4.50 – 5.00</td><td>Excellent (E)</td></tr> <tr> <td></td><td>4</td><td>3.50 – 4.49</td><td>Very Good (VG)</td></tr> <tr> <td></td><td>3</td><td>2.50 – 3.49</td><td>Good (G)</td></tr> <tr> <td></td><td>2</td><td>1.50 – 2.49</td><td>Fair (F)</td></tr> <tr> <td></td><td>1</td><td>1.00 – 1.49</td><td>Poor (P)</td></tr> </table>				Legend:	Scale	Numerical Rating	Descriptive Rating		5	4.50 – 5.00	Excellent (E)		4	3.50 – 4.49	Very Good (VG)		3	2.50 – 3.49	Good (G)		2	1.50 – 2.49	Fair (F)		1	1.00 – 1.49	Poor (P)
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Table 8 shows the results of the evaluation from the IT Experts for the development of PawPal: An Android Based Automated Pet Feeder for Pet Owners in Saint Joseph Village 10 Phase 1. The system garnered an overall weighted mean of “4.55” having an overall descriptive rating of “Excellent”.

The following are the results and findings gathered from the evaluation of the IT Experts:

1. **Functional Suitability:** The IT experts evaluated the system with a weighted mean of “4.56” having a descriptive rating of “Excellent” because of the system’s capability to perform required functions and features.
2. **Performance Efficiency:** The IT Experts evaluated the system and garnered a weighted mean of “4.6”, having a descriptive meaning of



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

“Excellent” due to the systems efficient implementation and performed the expected tasks.

3. Compatibility: The IT Experts evaluated the system and had a weighted mean of “4.65” with a descriptive rating of “Excellent” because of the systems capability to interact with both hardware and software and work seamlessly together.
4. Usability: The IT Experts evaluated the system and had a weighted mean of “4.45” with a descriptive rating of “Very Good” because the system promotes user-friendliness and has a very simple yet functional interface.
5. Reliability: The IT Experts evaluated the system and gathered a weighted mean of “4.55” with a descriptive rating of “Excellent” due to the systems capability to perform functions on a certain period of time and the capability of the hardware to keep up with the tasks.
6. Security: The IT Experts assessed the system and garnered a weighted mean of “4.52” with a descriptive rating of “Excellent” because the system is secure enough to handle sensitive information and promote data integrity.
7. Maintainability: The IT Experts gave the system a weighted mean of “4.44” and a descriptive rating of “Very Good” because the system's architecture is easy to maintain and perform well under given circumstances.
8. Portability: The IT Experts assessed the system and gave a weighted



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

mean of “4.7” with its descriptive meaning of “Excellent” because of the hardware’s convenience when it comes to size, proportions and weight

Chapter 5

SUMMARY OF FINDINGS, CONCLUSION, AND RECOMMENDATION

SUMMARY OF FINDINGS

With the test and assessments conducted for the capabilities of the proposed project, the following are the results gathered and findings of the study:

1. Functional Suitability - The respondents rated the proposed system as “Excellent” because the system performs the expected functions while meeting requirement standards which shows a successful implementation of the hardware and software functions.
2. Performance Efficiency - The respondents rated the proposed system as “Excellent” because the system met the expected performance capacity and that the system behaves accordingly.
3. Compatibility - The respondents rated the proposed system as “Excellent” because both hardware and software works well together and



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

is integrated properly. The software performs efficiently with no issues.

4. Usability - The respondents rated the proposed system as “Very Good” because of the system's ease of use, having a user-friendly interface and having a good interface aesthetics. The software is easy to learn and operate with no maximal efforts needed.
5. Reliability - The respondents rated the proposed system as “Excellent” due to its reliable operation and consistent performance throughout the testing period, with its quick response to faults and ability to execute functions under a given circumstance.
6. Security- The respondents rated the proposed system as “Excellent” because of its way of handling important information and promoting integrity of data throughout the entire system.
7. Maintainability - the respondents rated the proposed system as “Very Good” due to its simplicity and convenience to maintain. With the simple hardware and software functionality, the system is deemed to be easy to sustain.
8. Portability - the respondents rated the proposed system as “Excellent” because of the system's structure and construction, with the hardwares sleek and simple design, it can adapt to different environments well.



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

CONCLUSION

The following conclusions were derived based from the objectives of the study and result of the evaluation from the respondents.

1. The development of “Pawpal: an Android Based Automatic Pet Feeder for Pet Owners in Saint Joseph Village 10 Phase 1”.
2. The system is capable of providing an Android Based pet feeder for their pets.
3. The system is capable of dispensing dry food on specified measurement.
4. The system is capable of scheduling food plans using a mobile application with its given frequency.
5. The system is capable of giving real time status of the food container level.
6. The system is capable of providing feeding reports and feeding history of their pets.

RECOMMENDATIONS

Based on the results of the study, the following recommendations are made:

1. The system can include a camera monitoring system for a real time visual update on the pet.
2. The system should include portion selection for pets with special dietary



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

needs.

3. The system should have a pause and resume feature for scheduled meals.
4. The system should have a container wiper to remove leftover food from the pet.
5. The system can measure the amount of food in the container and displays it in percentage form.
6. The system can choose from four meal types: Breakfast, Lunch, Snack, and Dinner.
7. Each meal can be scheduled for a specific time, either in the AM or PM.



PAMANTASAN NG LUNGSOD NG MUNTINLUPA

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